From Data to Discovery: A Causal Discovery Pipeline for Biological Research

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Abstract

This work aims to critically evaluate a pipeline for Causal Discovery. By leveraging observational data, the pipeline is able to identify causal relationships and the counterfactuals that are able to be derived from observational data. A Python implementation of the pipeline is provided and demonstrated.

1 Introduction

Traditional data analysis methods predominantly rely on observational data, which imposes limits regarding the identifiability of causal structures. Interventional data helps establish a cause-and-effect relationship by breaking the influence of confounding variables. In some cases, counterfactuals pertaining to interventional data can be derived from just observational data.

This process of determining whether and how one variable affects another variable, beyond correlation, is called Causal Inference. It should precede effect estimation, as bayesian statistics rely on prior assumptions about the relationships in the data. Only by combining Causal Inference and modern Machine Learning techniques can we arrive at conclusions about cause and effect, answering questions such as "What would happen if a variable had been different?".

Inverse problems, with the goal of understanding the causal factors that produced a set of observations, are a complex challenge in many fields, biology naturally being one of them. This work will focus on Causal Discovery, finding the underlying causal structures between variables, meaning it will not estimate these effects, which is traditionally done with methods like Regression. For this purpose a pipeline for Causal Discovery and counterfactual discovery using Python is developed and tested on synthetic and real-world data.

The theoretical underlying model, modeled as a graph, which accurately describes the causal relationships between the variables which can be observed is called the true structural causal model (true SCM)[1]. We want to partially recreate and infer information from the true model. Furthermore, from the recreated SCM we want to determine mathematical expressions using Judea Pearls Do-Calculus[2] to describe the causal effect. Therefore the pipeline is composed of the following steps, with each step outputting the next steps necessary input3:

1. Identification of Conditional Independencies within the dataset
2. Learning the Causal Graph (PC, FCI)

3. Refine Graph (Incorporate prior knowledge)
4. Identify Causal Effects (do-calculus[2], ID, IDP)

This pipeline will yield discoverable counterfactuals, given the assumptions, and it's based on the approach for specifying causes developed by Pearl and Woodward.

For this pipeline, the necessary requirements and assumptions will be explained, justified and critically evaluated to answer these guiding questions:

1. How can interventional data be inferred from observational data?
2. What are the requirements and assumptions to infer interventional data from observational data?
3. What are the challenges of Causal Discovery in real data?
4. Which algorithms for each step will lead to the best results under what conditions?
5. How can additional knowledge be used to achieve better results?
6. What factors, such as dataset size, affect the success of Causal Discovery?

For each step and algorithm this workstrategies from research based on the work of Judea Pearl, as other recent research regarding explainable AI has raised well-founded questions regarding the validity of other algorithms used to determine counterfactuals.[3]

2 Preliminaries

Attribute substitution is the process, in which a target judgment is effortlessly and automatically made by substituting a simpler, natural judgment.[4] This is often applied in real life, when the question for causality is replaced by that for correlation. Correlation between two variables does not imply a causal relationship of these variables, and neither does a causal relationship imply correlation. But not only that, no correlation between two variables does not imply no causal relationship, and neither does no causal relationship of variables indicate no correlation.

A causal effect of X on Y is defined through an intervention $do(X = x)$[2], leading to the interventional distribution $P(Y \mid do(X = x))$. A causal relationship is established if:

$$P(Y \mid do(X = x)) \neq P(Y) \tag{1}$$

This notation reads: The distribution of Y after intervening to set $X = x$ is not equal to the distribution of Y. This implies that actively setting X to a fixed value changes the distribution on Y, whichs means that X has a causal effect on Y. Y is commonly called the outcome, and X the treatment.2.6

Correlation without causation may be the result of a common cause or a spurious relationship. Causation without correlation may appear when there are unobserved compensatory effects, a form of saturation has been reached that diminishes the causal effect, only a small range of possible values is observed or the variables have a nonlinear causal relationship.

Table 1: Examples of Relationships Between Correlation and Causation

X **and** Y	**Causation**	**No Causation**
Correlation	• Smoking → Lung Cancer[5] • Sun Exposure → Skin Cancer[6]	• Ice Cream Sales ↔ Drowning[7] • Air pollution in Oklahoma City ↔ Google searches for 'funny cat videos'
No Correlation	• Quadratic Relationship: $Y = X^2$ • Absolute Value: $Y = \|X\|$	• Shoe Size ⇹ IQ • Random Variables

Note: Arrows (→) indicates causal direction, double arrows (↔) indicate correlation without causation.

2.1 Structural Causal Model

A *structural causal model (SCM)* is the formal framework in causal science and the basic semantical framework[1], and what is assumed to be able to display the true underlying data. It represents causal relationships using structural equations, which can be modeled as directed graphs. The SCM is defined as a tuple $\langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P(\mathbf{U}) \rangle$, where:

- $\mathbf{U}$: A set of *exogenous variables* whose values are determined by factors outside the model.
- $\mathbf{V}$: A set of *endogenous variables* whose values are determined by factors within the model.
- $\mathbf{F}$: A set of *structural equations* that express each endogenous variable $V_i \in \mathbf{V}$ as a function f_i of its parent variables $\mathrm{Pa}(V_i)$ and exogenous variables:
$$V_i = f_i(\mathrm{Pa}(V_i), U_i), \quad \text{for } i = 1, 2, \ldots, n$$
- $P(\mathbf{U})$: A joint probability distribution over the exogenous variables, representing the error terms.

The *structural causal model* differs from a causal model in it's distinct equations that each variable has to have.[1, p. 27] In this work and the resulting program SCMs are used to model the true models, as SCMs can be parametric or non-parametric. The SCM does not assume the independence of exogenous variables.[8] The causal relationships defining the SCM can be represented by a directed acyclic graph.

2.2 Directed Acyclic Graphs and Partial Ancestral Graphs

A directed acyclic graphs (DAG) is a graph representations of the causal relationships and the assumed data-generation process of a set of variables. It

formally consist of nodes and directed edges with the condition of Acyclicity. Though it represents the causal relationships of a SCM, it does not further specify strength, sign and whether the relationship is functionally linear or non-linear.

Similarly the *Partial Ancestral Graph (PAG)* is a graph representation of causal relationships of a *Markov equivalence class (MEC)*, which is the collection of causal graphs that share the same set of observed features"[9, p. 1]. It allows for edge-endings to be unidentified, meaning a an arrowhead or tail. This uncertainty is visualised as a circle at the end of an edge.

2.3 D-Separation

D-separation is a core concept of causal discovery algorithms. A set of edges Z d-separates nodes X and nodes Y if and only if Z blocks every path, meaning any consecutive sequence of edges, between a node in X and a node in Y. It is denoted by

$$X \perp\!\!\!\perp Y \mid Z[1]$$

If Z d-separates X and Y, conditioning on Z, by, for example, setting a range on Z, blocks the path that could allow information to flow through. It makes X and Y conditionally independent, assuming they were to begin with. If, after conditioning on Z, conditional dependence between X and Y persists, it suggest a causal relationship between the variables or unmeasured confounding. As such, d-separation encodes independence assumptions in causal graphs.

On PAGs, the similarly defined, slightly modified version of d-separation, which also encodes the conditional independencies, is called *m-separation*. For the same reason, to be applicable to PAGs, as the existence of m-separation the semi-markov assumption and m-faithfulness exist.[8]

2.4 Confounding

When variables influence both treatment and outcome variables, which introduces bias to the data, they are called *confounders*.[1] The conditioning on confounders is not sufficient to eliminate bias introduced by confounders, as complex causal structures cause bias to find another way, when one is blocked. Confounders that cannot be measured are called *latent confounders*. In 2.5 $X2$ is a confounder between $X3$ and $X4$.

2.5 Collider

A variable that is influenced by treatment and outcome is a *collider* on the path between them. Conditioning on a collider may introduce selection bias through spurious association between treatment and outcome, even if they are conditionally independent.[10] In 2.5 X3 is a collider on the path from X1 to X2.

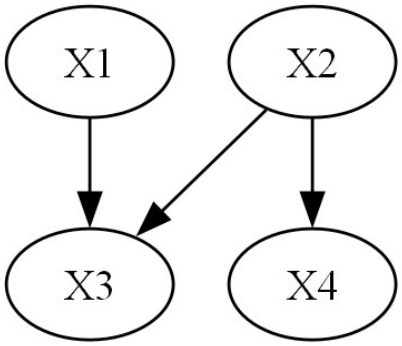

Figure 1: A simple DAG representing a SCM

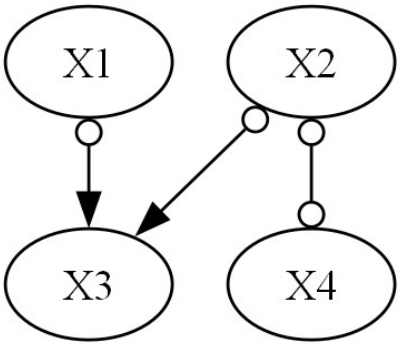

Figure 2: A derived DAG (using FCI) representing a MEC

2.6 Do-Calculus

The do-calculus provides the rules to estimate interventional distributions from observational data using causal graphs. *"There is no way a model can predict the effect of an action unless one specifies what variables in the model are directly affected by the action, and how. The do(x) operator is a mathematical device that helps us specify formally what is held constant, and what is free to vary in any given experiment."*[1, p. 358]

The rules of the do-calculus are complete, meaning that if it is possible to identify the causal effect, the rules of do-calculus can be applied to transform the initial "causal effect formula into a formula that only includes observational quantities"[11]1

The completeness of the do-calculus also applies to the ID[12] and IDP[9] algorithms, which are used in the pipeline.

3 Theoretical foundation of the pipeline

The objective of the pipeline is to infer mathematical expressions for causal relationships from observational data. To achieve this the pipeline will first calculate the conditional independencies to generate a causal graph using the FCI[13] or PC algorithm[14]. Given the assumptions this generated causal graph will have a subset of edges and edge orientations of the true causal graph, or circles, equivalent to a *Markov equivalence class* (MEC)3. It will not contain edges that are not in the true causal graph, and no edge that is orientated in one direction in the true causal graph will be directed in a different direction in the generated causal graph. On this graph we will use the ID[12] algorithm, which will return mathematical expressions for the causal relationship of the variables in the observed dataset.4

Given the assumptions, the first step cannot give false results (Markov Assumption and Faithfulness). It returns a list of the all and only the independen-

cies that the true SCM has as well. The FCI algorithm is complete [15]. The ID Algorithm is complete [12]. The IDP algorithm is complete. [9] That means, given the assumptions, the pipeline will return all results that can possibly be derived from the observed data. When applied practically however, the Markov assumption and the Faithfulness assumption may be broken. This can have three potential reasons, measurement noise leading to false positives or negatives, too low sample size to draw significant conclusions and incompleteness of the calculation of conditional independencies. That means we need to optimize this first step of the pipeline, the identification of conditional independencies within the dataset, to ensure we get the most accurate results.

Regardless, "the algorithms can produce correct output even though some required statistical decisions are made incorrectly".[13, p. 131]

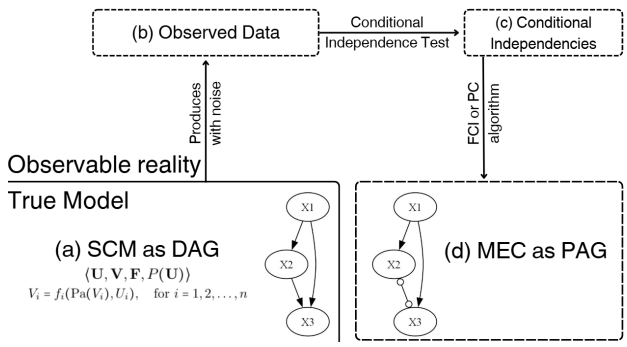

Figure 3: Note: FCI is applied here. PC produces a DAG instead.

4 Assumptions

The suggested approach underlies the limitations of its parts, and as such, assumptions about the true SCM and the measurements have to be made. These are the keys assumptions for the entire pipeline:

1. Acyclicity: The true SCM is acyclic. This assumption is necessary to learn the Causal Graph with the PC[14] and FCI[13] algorithms.

2. Markov Assumption: Each variable is conditionally independent of its non-descendants given its parent nodes. Without this assumption it is not possible to simplify complex the complex causal structures into sets of conditional independence statements. So for each conditional independency in the true distribution P and the identified conditional dependencies in graph G it holds that:

$$X \perp\!\!\!\perp Y_G \mid Z \implies X \perp\!\!\!\perp Y_P \mid Z$$

3. Faithfulness: Statistical dependencies correspond to d-seperations in the underlying causal graph. The observed data has the same conditional dependencies as the true causal model. If this is violated, the algorithms to learn causal graphs can return graphs with inversely oriented or missing edges, compared to the true causal models graph.

$$X \perp\!\!\!\perp Y_G \mid Z \Longleftarrow X \perp\!\!\!\perp Y_P \mid Z$$

4. Causal Sufficiency: All common causes are observed and included in the data (not necessary for FCI).

These assumptions can be summarized: The underlying SCM is acyclic, the conditional dependencies calculated in the first step of the pipeline are the exact same as those of the SCM, and for PC additionally all common causes must be observed and included in the data. Given these assumptions, the pipeline will produce counterfactuals that can be identified from observational data, or return nothing if the causal effect is not identifiable.

5 Identification of Conditional Independencies

The field of Causality relies on statistical tests of independence to determine the relationships between variables. In complex systems the choice of statistical independence test method affects the outcome significantly. The following sections will briefly underline the significance of the conditional independence testing in context of the pipeline, and then summarize the statistical tests provided by the causal-learn library[16], the backbone of the causality pipeline, and provides guidelines for deciding which statistical test to utilize.

5.1 Significance of the Conditional Independency Test

As the PC and FCI, and the ID and IDP Algorithm are complete, errors can only occur in the independence testing section of the pipeline. Should such an error occur, then it may violate the Faithfulness-, as well as the Markov-, assumption. Identifying conditional independence relationships forms the foundation of Causal Inference. Because this is the only vulnerable part of the pipeline, the choice of which statistical dependence test to use should be well considered, as each test has advantages, as well as disadvantages, which this work aims to explain.

5.2 Fisher-Z Test

The Fisher-Z test is one of the most widely used conditional independence tests, because it yields optimal results for linear-Gaussian data.[17] It evaluates whether two variables X and Y are conditionally independent given a set of variables $\mathbf{Z}$. The algorithm makes use of the Pearson correlation coefficient to transform it into a normally distributed variable for hypothesis testing.

The Fisher-Z transformation is defined as:

$$z_r = \frac{\ln\left(\frac{1+r}{1-r}\right)}{2},$$

where r is the correlation coefficient.

This is done by comparing the transformed value z_r to the standard normal distribution under the null hypothesis of conditional independence. The test calculates a correlation coefficient $Z \in [-1, 1]$, which reflects directionality in the relationship. This approximation improves in accuracy as dataset size increases.

One assumption the Fisher-Z test makes and that limits it's applicability in real world scenarios is, that the variables are normally distributed. When normality is not given, increased sample size may not improve results at all, it might even worsen them, a study from 2016, testing Fisher-Z tests on nonnormal data and comparing the confidence intervals with Monte Carlo simulation, found. [18].

When the conditioning set is large relative to the sample size, the Fisher-Z transformation can lead to biased and inconsistent estimates.

The reliance on Pearson correlation coefficients further limit the Fisher-Z tests applicability on real world data. The Fisher-Z test systematically fails to detect non-linear relationships, regardless of sample size. In fields where non-linear relationships are expected, such as gene networks or socioeconomic systems, results should be interpreted with caution.

How much caution is required is a debated topic.[19] Because of it's accuracy when dealing with linear relationships in Gaussian-distributed data and the common occurence of those it is still a very popular testing tool.

5.3 Chi-Square Test

The chi-squared test of independence (also χ^2 test) is a non-parametric test that examines relationships between variables, as such it serves as a fundamental tool for discrete variables. The test calculates whether the observed joint distribution of two variables differs significantly from what would be expected, were they independent. Especially for categorical data analysis this is the classical approach. The chi-squared test is used only to test the presence of an association between two categorical variables, it does not determine or assume directionality.

The test statistic is calculated as

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

where O represents observed frequencies and E represents expected frequencies. [20] This follows a chi-squared distribution under the null hypothesis that the variables are independent. The test evaluates whether this observed joint distribution of two variables differs significantly from what would be expected if they were independent.

> Non-parametric tests should be used when any one of the following conditions pertains to the data:
>
> 1. The level of measurement of all the variables is nominal or ordinal.
> 2. The sample sizes of the study groups are unequal. For the χ^2 test, the groups may be of equal or unequal size, whereas some parametric tests require groups of equal or approximately equal size.
>
> [20, p. 143]

Because the chi-squared test is robust to various skewness and, unlike the parametric tests like the Fisher-Z test, works for categorical data, it became a very popular testing method. The calculations this tests required are also very efficient.

However, the chi-squared test requires the discretization of continous variables, which may introduce bias and lose information. For the chi-Squared test, the size of expectations E should be large, many writers recommend that it should not be less than 5. [21, p. 418] When the assumption of sufficiency is violated, the test statistic no longer follows a chi-squared distribution under the null hypothesis, leading to incorrect p-values and flawed inference. The curse of dimensionality also applies to the chi-squared test: As the conditioning set becomes larger, the required sample size increases dramatically.[22]

5.4 G-squared Test

The G-Squared test, also known as the likelihood ratio test, is an alternative to the chi-square test for assessing independence between categorical variables. It is computed using the formula:

$$G^2 = 2 \sum O \ln \left(\frac{O}{E} \right)$$

.

Just like in the chi-squared test, the O represents observed frequencies and E represents expected frequencies. The statistic also follows a chi-squared distribution under the assumption of independence. Thus, the G-squared test generally performs similarly to the chi-squared test, though it may have different results when working with small sample sizes.

Also like the chi-squared test, discretization of the data is required. If the data is grouped into a small number of categories, the information loss may be bigger, this can inhibit the ability to detect relationships. However, the categories should each be as large as possible, each cell counting at least 5 elements to achieve valid results.[21, p. 418] This fundamental tradeoff has implications on the ease of use of the chi-squared and G-squared test, as well as the setup needed to produce reliable results. Generally, no discretization approach can fully preserve the information in the original continuous data.

While the chi-squared test of independence calculates a test statistic and based on squared differences between observed and expected frequencies, the G-squared test measures the deviation between observed and expected frequencies on a logarithmic scale. On small samples, the G-squared test tends to yield more accurate results than the chi-squared test. But generally, both tests can be unreliable when expected frequencies are low.

5.5 Kernel-based Conditional Independence Test

The Kernel-based Conditional Independence test (KCI-test), developed by researchers at the Max Planck Institute for Intelligent Systems in Germany, is a non-parametric conditional independence test. In this test, Kernel Hilbert spaces are used to infer conditional independence without making assumptions about the true distribution.[22] Through the use of a higher-dimensional feature space, linear methods can identify non-linear relationships in the original space. As the KCI-test was developed with the goal of advancing Causal Inference, it exhibits strong performance across various scenarios when working with data containing non-linear dependencies and mixed variable types.[22] Compared to the Fisher-Z and Chi-Squared test, it is not as efficient, taking many times longer to calculate the conditional independencies. It scales at least quadratically with sample size.[23]

5.6 Summary of Independence Tests

The range of tests described are versatile enough to handle diverse data types. There are adaptations, which have no yet been widely implemented, that specialise usage further. There exist many varieties of the Fisher-Z test of conditional independence, aiming to improve accuracy in specific scenarios.[24] There also exist approximate KCI-tests for conditional independence.[23] However, the causal-learn library does not currently implement those.[16]

6 PC and FCI algorithms

6.1 PC

The *Spirtes, Glymour and Scheines* algorithm (SGS)[25] and soon after the *Peter-Clarke* algorithm revolutionized the field of Causality with a reliable, complete method for causal discovery with causal sufficiency. Given the assumptions 4, these algorithms use the contraints, the conditional independencies, found on the true structural model to reconstruct the DAG modeling the true SCM.[13, p. 84-85] The use of reliable conditional independence testing methods is required.[8]

When applied to data, the PC algorithm[25] first calculates the conditional independencies in the data. It then constructs a complete graph with undirected edges and remove any edges connecting conditionally independent nodes. By further exploiting the found conditional independencies and recursively applying

Statistical Independence Test	**Advantages**	**Disadvantages**
Fisher-Z Test	Optimal for Gaussian data; computationally efficient; accurate for linear relationships; good asymptotic properties	Requires normality assumption; only reliably detects linear relationships; doesn't work with categorical variables; requires large sample size; poor performance with many conditioning variables; significant outliers can strongly influence the result
Missing-Value Fisher-Z Test	Can handle incomplete data and preserves more samples than deletion	Poor performance on low sample size; missing pattern correlates with variables of interest; high percentage of missing values can distort the result; less computationally efficient than Fisher-Z
G-squared/Chi-Squared Test	No distributional assumptions; naturally handles categorical data; robust to certain types of outliers	Does not work on continuous data without discretization; requires high expected cell counts; detrimental at high cardinality
Kernel-based Conditional Independence Test	Detects non-linear dependencies; Handles mixed variable types	Long computation time makes data with high sample size difficult to process

Table 2: Comparison of Independence Tests for Causal Discovery

Test	**High Dimensionality**	**Nonlinear Data**
Fisher-Z Test	Large sample size needed	Assumes linearity
Missing Value Fisher-Z	Requires larger sample size	Assumes linearity
Kernel CI Test	Handles high dim. via kernel trick	Accurately models non-linearity
G-Squared Test	Not accurate for high dim.	Some nonlinearity handled
Chi-Square Test	Not accurate for high dim.	Requires categorical data, poor non-linearity

Table 3: Comparison of statistical independence tests under various data conditions.

transformation rules it attempts to reconstruct as much of the true SCMs DAG representation. Because the calculated conditional independencies represent constraints on the graph[15] the PC algorithm is commonly called a *constraint-based causal discovery method*[16].

6.2 FCI

If causal sufficiency is not given, meaning there may exist unobserved confounders, the PC algorithm does not perform optimally. *The Fast Causal Inference (FCI)*[13, p. 138-145] algorithm also reconstructs a causal graph from observational data[13], however is complete even without causal sufficiency. Unlike the PC algorithm it produces a MEC in the form of a PAG, to accurately display the unknown status of some edge endings.2.5 Given causal sufficiency, PC generally performs better than FCI. But as a more generally applicable algorithm the FCI "is feasible in large variable sets provided the true graph is spare and there are not many bidirected edges chained together"[13, p. 129]

Given faithfulness the FCI will produce a partially oriented inducing path graph that contains no edges, that the true DAG representing the SCM also did not contain. There may be uncertainty in the directionality, but the structure will be a a subset of to the true DAGs structure.

The FCI algorithm[13, p. 138-145] works similarly to the PC algorithm, in that it first calculates conditional independencies and then reconstructs the graph as accurate as possible. Because it has to account for the existence of latent confounding between variables, in cases where causal sufficiency is given, it will not decide the directionality of some edges that PC does decide.

Score-based test	**Input**	**Output**	**Causal Sufficiency required**
PC	Data	DAG2.2	Yes
FCI	Data	PAG2.2	No

Table 4: Overview of score-based tests used in pipeline

6.3 ID and IDP algorithms

To quantify the effect of counterfactuals mathematical expressions pertaining to the intervention have to be derived. Asking what would happen to a variable Y when intervening to set $X{=}x$, given DAG G, is a complicated question. The task of doing so is known as **conditional effect identification.**[26] While the do-calculus2.6 provides the mathematical framework and the theoretical guarantee, that any discoverable counterfactuals in a DAG can be derived using consecutive application of the do-calculus rules, it does not provide a algorithmic mean of calculating this consecutive application.

The ID algorithm[27] provides a complete identification algorithm, based on the complete graphical criterion for identification of conditional interventional distributions. A DAG is required as input, which is divided The IDP algorithm provides this same Conditional Effect Identification, but requiring a PAG as input. Both the ID and the IDP algorithms are complete.[27, p. 4] [28, p. 6]

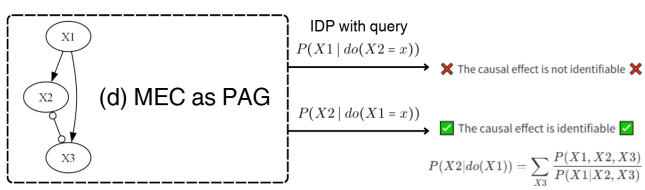

Figure 4: Note: Same true SCM as in 3.

7 How to use the pipeline

The code for the implementation of the pipeline, using the causal-learn package in Python[16]and the PAGId package in R[26], authored by Dr. Adèle Ribeiro. It can be downloaded from Github, where a documentation can also be found.

Though the IDP feature could not be made available on the Docker version yet, Docker can be used to effortlessly run the program in a controlled environment, without having to download the dependencies individually, and still utilize FCI with background knowledge. To run the pipeline implementation with IDP analysis, it is required to download the dependencies.

The data input into the pipeline is converted locally. Because some of the algorithms used are not efficient, especially for large sample sizes and feature spaces, substantial processing power may be required.

After starting the program, data can either be input or randomly generated, to test the functions of the pipeline. The generated data follows gaussian distributions, with randomly generated causal relationships between the variables. When running either PC or FCI algorithm, there are options to also set the conditional independence test type and p-value used in the calculation of the conditional independencies.

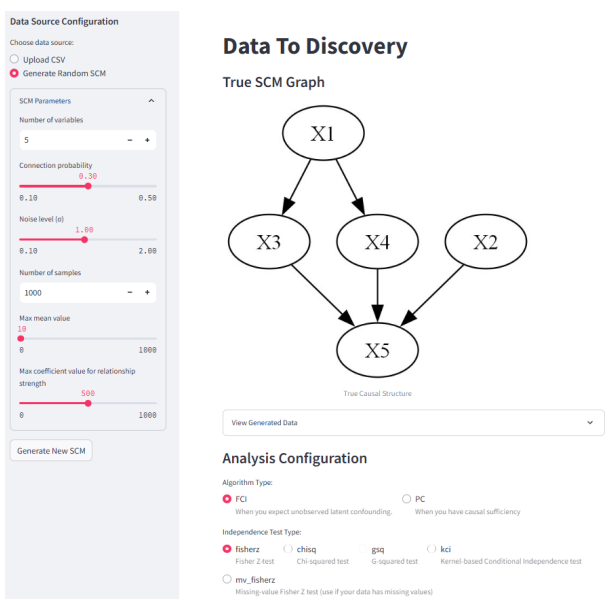

Figure 5: The GUI of the pipeline

7.1 Background Knowledge

The knowledge of already existing causal relationships in data can ease the identification of other causal relationships. Using this knowledge, more accurate

and extensive conclusions about the true SCM can be inferred. For example, age and sex are generally not outcomes of a target variable in data, meaning that no other variable causes age or sex. Using a tier map accessed in the GUI, it is possible to pass this background knowledge to the algorithm, which it will use in the calculations to avoid causal structures that include age and sex as descendants of other variables. When a causal relationship is either already known or known to be absent, orientating the corresponding edge or forbidding it using background knowledge can cause chain reactions aligning other unknown edges.

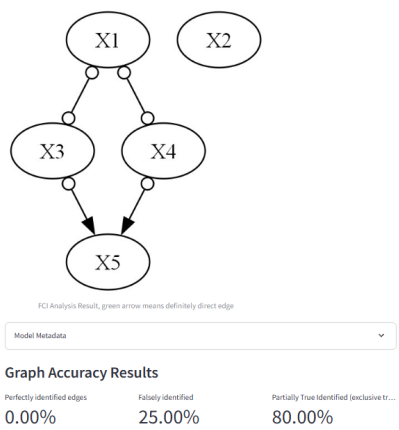

Figure 6: No background knowledge is applied

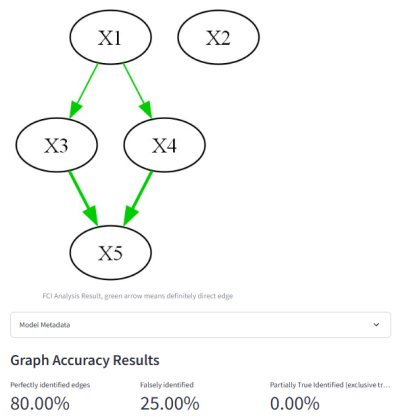

Figure 7: Background knowledge $\text{tier}(X_1) > \text{tier}(X_n)$, where $n \in \{2, 3, 4, 5\}$, applied

The graph in 5 represents the SCM that generated the data used when calculating 7.1 and 7.1.

The causal-learn libraries[16] implementation of tiered background knowledge has been altered for the pipeline. Instead of applying tiers only to select variables and ensuring that edges between them only possess one causal direction, every variable is associated to a tier. This choice was made because it covers the case that some variables are not outcomes of others at all more extensively. In the original implementation it was impossible to pass background knowledge in a way that one node could have connections to all others without effecting the others' relationships as causal relationships within a tier were forbidden. That means the original behaves differently when all nodes are in the same tier and when an empty background knowledge is passed. This opposite is true for the pipeline implementation.

7.2 Evaluating algorithm performance in controlled scenarios

When using generated data, the underlying SCM is known. Thus the accuracy can be rated. In this work, the *percentage of perfectly identified edges* represents edges with the same properties in both the DAG representation of the true SCM and the algorithmically calculated PAG, meaning directed edges without circles. The *percentage of false identified edges* represents the ratio of missing, falsely oriented or falsely added edges to partially true identified edges. The *percentage of partially true identified edges* represents the ratio of edges that exist in the true SCMs DAG, that have circles in the evaluated PAG, to the edges in the true SCM. It presents a way of quantifying the performance of the discussed algorithms.7.1

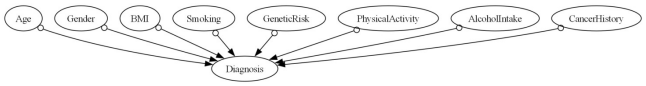

Figure 8: Result of FCI applied on a synthetic dataset[29] containing commonly observed causes of cancer without applied background knowledge. As expected, the graph partially identifies the edges, implying that all variables may be causes of the diagnosis

8 Application of the pipeline

Applying the implementation of the pipeline can generally requires preprocessing the data, due to it only performing well when missing values using the missing-value Fisher-Z test and this does not work well with mixed nor non-linear data2. The KCI-test requires can appropriately process mixed variable types and assess , but is inefficient and missing-values may introduce bias.

Using a preprocessed version[30] of a dataset created in 1988 by merging four databases from Cleveland, Hungary, Switzerland, and Long Beach VA,[31] the process of applying the pipeline will be demonstrated. Because the data includes age and sex, which cannot be outcome variables in our data, we can apply the background knowledge

$$\text{tier}(\text{age}) = \text{tier}(\text{sex}) < \text{tier}(n) \quad \forall n \notin \{\text{age}, \text{sex}\}$$

After loading the data and applying the specified background knowledge and using the Fisher-Z test of independence, a DAG visualising the graphs MEC is output. The Fisher-Z test was used because of constraints on computational power, the KCI-test may deliver better results, especially because the dataset consists of mixed variables.

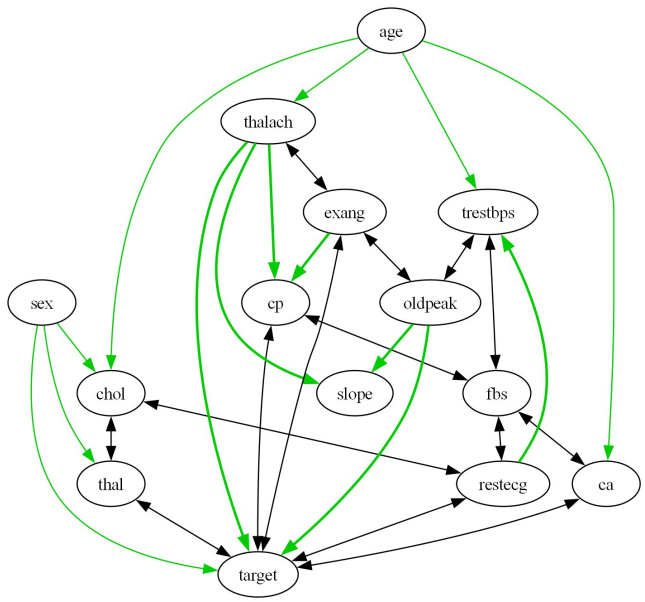

Figure 9: Result of FCI on the Heart Disease Dataset[31] using background knowledge: sex and age cannot be descendants of other nodes

9 Results and Discussion

The result of the FCI algorithm implies a causal relationship between the maximum achieved heart rate ("thalach") and the presence of heart disease in the patient ("target"). A link between these variables has been studied[32] and this outcome result servers as further evidence. At this stage it is stil unclear, whether this causal effect can be identified, meaning the distribution of the *marginal effect*[26] of the maximum achieved heart rate on heart disease can be identified given the PAG. In that case that it wasn't identifiable, the interventional distribution could not be mathematically expressed.

Using the IDP algorithm the causal effect is evaluated to be identifiable10. The resulting formula can further be used to mathematically calculate an accurate mathematical term for any intervening $x \in$ Thalach.

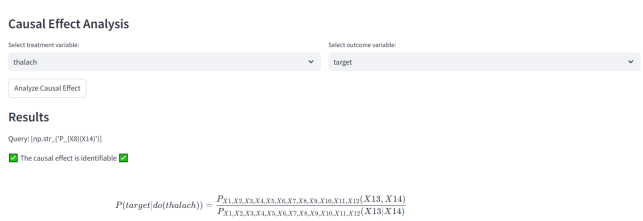

Figure 10: Excerpt of the resulting expression of $P(target \mid do(thalach = x))$

The PAG representing the FCI result significantly differs when not applying background knowledge.11 The graph now implies an effect of restion blood pressure on age, among other fallacies. The IDP still evaluates the marginal causal effect as identifiable, with the resulting expression also differing from the expression derived from IDP when applying background knowledge to the conditional independence testing and FCI.

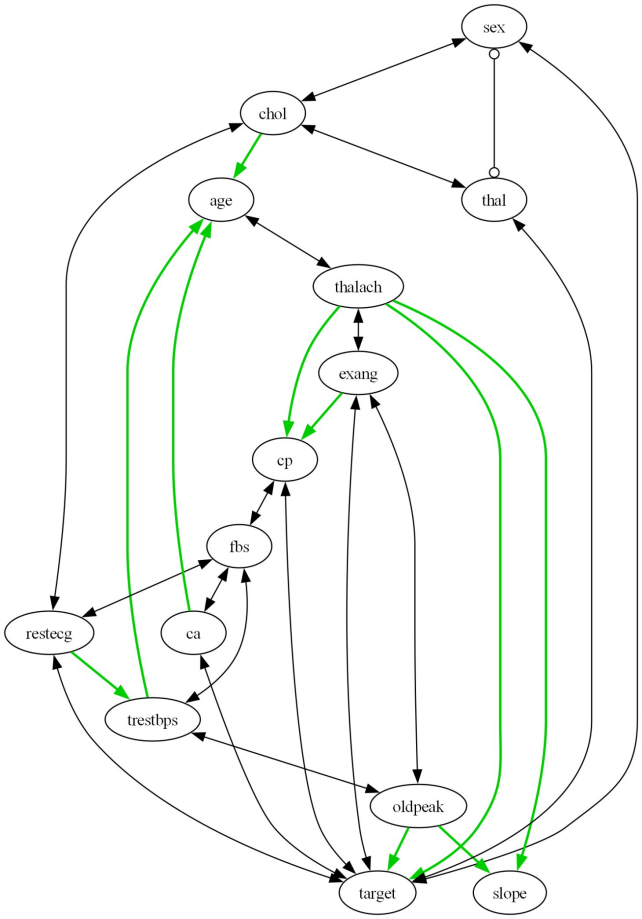

Figure 11: Result of FCI on the Heart Disease Dataset[31] without using background knowledge

With and without background knowledge the algorithm is prone to making erroneous conclusions, but the issue is exacerbated when not applying background knowledge. The inclusion of various types of background knowledge can make the result of constraint-based causal discovery methods more accurate. Thus, the researcher should be actively engaged in the Causal Discovery process, emphasizing the need for a user-friendly program.

Because the results of this pipeline are created without regard for the assumptions, they should be interpreted with caution. Causal Discovery using this pipeline is a means of gathering evidence regarding the causal relationship rather than providing definitive proof and it should be utilized as such by using it in conjunction with other data analysis tool, best proceeding tools for effect estimation like Regression. For this purpose Causal Discovery tools have to become easily applicable to become more relevant for data analysis.

10 Expectations and Impact

This research aims to advance the practical use of causal discovery methods by providing an open-source comprehensive pipeline and a Python-based tool to the scientific community. This work seeks to explore the conditions, under which causal discovery methods produce reliable results. Through provision of accessible data analysis tools with a user-friendly GUI this work hopes to contribute to the adoption of Causal Inference methods for practical scenarios.

Furthermore, the potential impact of Causal Discovery on the field of Biology is immense. The ability to derive counterfactuals based on interventional data from observational data could mean being able to predict the effects of medicine or treatments, before they are administered. It could provide valuable insights into complex biological systems, such as the workings of the brain and the mechanisms behind diseases.

Due to the difficulty of applying these algorithms and the severity of the necessary assumptions, the discussed algorithms based on the approach for specifying causes developed by Pearl and Woodward are scarcely utilized as a scientific method. But as the recent development of Machine Learning algorithms demonstrated, the most revolutionary methods may already be developed, but underutilized.

11 Reflection

The development process of the pipeline implementation involved iterative improvement of implementation strategies as well as architecture. Python was chosen as the primary programming language because the most popular algorithms commonly used for causal discovery have well-documented implementations (e.g. causal-learn[16]) and the extensive and well-maintained ecosystem to visualize information and create interactive graphical user interfaces. However, the only library that implements the IDP algorithm is exclusively available in R, the PAGId package[26]. That's why internally the current implementation of the pipeline necessitated data serialization between Python and R, introducing unanticipated complications. Because of computationally intensive code the debugging proved difficult, as errors could originate in either language, in any layer, and single program runs could take minutes. Because the pipeline is using algorithms from not only different packages but also languages, the list of

necessary dependencies to execute the Python program is also extensive.

In hindsight, a better approach would have been to first reimplement the IDP algorithm in Python or depend entirely on R, to eliminate the need for multiple programming languages.

12 Outlook

The current implementation of the pipeline provides the basic functionality to analyse data for causal relationships through a GUI. There are many ways to improve the performance, applicability and stability of the pipeline implementation. Preprocessing data requires human effort, and can be another cause of error, thus a way to automatically preprocess data appropriately can be provided. As there are not only many conditional independence tests, but also discretization methods for working with Bayesian Networks[33], the analysis options could be broadened to include more variety therein. Efficient approximation of the KCI-test[23] that are faster can make KCI more easily applicable with large dataset as the KCI-test performs best on mixed variables out of the discussed conditional dependency tests.5.5

Furthermore, to identify non-marginal effects the pipeline is sufficient, but the PAGId package also includes implementations of the CIDP algorithm for Conditional Effect identification.[26] Thus the implementation of the pipeline is a first effort in the direction of more widespread usage of Causal Discovery methods for data analysis tasks, with a lot of improvement potential, but provides basic functionality.

"It remains an open challenge to estimate the resulting causal expressions from finite samples."[34]

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